Integration by Parts

Section 1: Standard Products (Polynomial × Exponential / Trig)

1 State the formula for integration by parts.

a. $\int u\,dv = uv - \int v\,du$

b. $\int u\,dv = uv + \int v\,du$

c. $\int u\,dv = u'v - uv'$

d. $\int u\,dv = \frac{u}{v} - \int u'v$

2 Evaluate $\int xe^{2x}\,dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$

b. $\frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{4}xe^{2x} - \frac{1}{4}e^{2x} + C$

3 Evaluate $\int x\sin(x)\,dx$.

a. $-x\cos x + \sin x + C$

b. $x\cos x - \sin x + C$

c. $-x\cos x - \sin x + C$

d. $x\sin x + \cos x + C$

Section 2: Logarithmic and Inverse Trig Integrands

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4 Evaluate $\int \ln(x)\,dx$.

a. $x\ln x - x + C$

b. $\frac{1}{x} + C$

c. $x\ln x + x + C$

d. $\frac{(\ln x)^2}{2} + C$

5 Evaluate $\int \arctan(x)\,dx$.

a. $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$

b. $x\arctan x - \ln(1+x^2) + C$

c. $\frac{1}{1+x^2} + C$

d. $x\arctan x + \frac{1}{2}\ln(1+x^2) + C$

6 Evaluate $\int x\ln(x)\,dx$.

a. $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$

b. $\frac{x^2}{2}\ln x - \frac{x^2}{2} + C$

c. $x^2\ln x - x^2 + C$

d. $\frac{x^2}{2}\ln x + \frac{x^2}{4} + C$

Section 3: Repeated Integration and Tabular Method

7 Evaluate $\int x^2 e^x\,dx$.

a. $x^2e^x - 2xe^x + 2e^x + C$

b. $x^2e^x - 2xe^x + C$

c. $x^2e^x + 2xe^x + 2e^x + C$

d. $\frac{x^3}{3}e^x + C$

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8 Evaluate $\int x^2 \cos(x)\, dx$.

a. $x^2 \sin x + 2x \cos x - 2 \sin x + C$

b. $x^2 \sin x - 2x \cos x + 2 \sin x + C$

c. $-x^2 \cos x + 2x \sin x + C$

d. $x^2 \sin x + 2x \cos x + C$

9 Evaluate $\int_0^1 x^2 e^{2x}\, dx$.

a. $\frac{e^2 - 1}{4}$

b. $\frac{e^2}{4}$

c. $\frac{e^2 + 1}{4}$

d. $e^2 - 1$

Section 4: Looping / Cyclic Integrals and Definite Bounds

10 Evaluate $\int e^x \cos(x)\, dx$.

a. $\frac{1}{2} e^x (\cos x + \sin x) + C$

b. $\frac{1}{2} e^x (\cos x - \sin x) + C$

c. $e^x (\sin x + \cos x) + C$

d. $\frac{1}{2} e^x \sin x + C$

11 Evaluate $\int_1^e \ln(x)\, dx$.

a. 1

b. $e - 1$

c. e

d. 0

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12 Evaluate $\int_0^{\pi} x\sin(2x)\,dx$.

a. $-\pi/2$

b. $\pi/2$

c. 0

d. $-\pi$

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